

tf.broadcast_to Stay organized with collections Save and categorize content based on your preferences.

https://www.tensorflow.org/api_docs/python/tf/broadcast_to

Broadcast an array for a compatible shape.

Function Signatures

```
tf.broadcast_to( input: Annotated[Any, TV_BroadcastTo_T],
  shape: Annotated[Any, TV_BroadcastTo_Tidx], name=None ) ->
  Annotated[Any, TV_BroadcastTo_T]
x = tf.constant([[1, 2, 3]]) # Shape (1, 3,)
y = tf.broadcast_to(x, [2, 3])
print(y)
```

Code Examples

```
tf.broadcast_to(  
input: Annotated[Any, TV_BroadcastTo_T],  
shape: Annotated[Any, TV_BroadcastTo_Tidx],  
name=None  
) -> Annotated[Any, TV_BroadcastTo_T]
```

```
tf.broadcast_to(  
input: Annotated[Any, TV_BroadcastTo_T],  
shape: Annotated[Any, TV_BroadcastTo_Tidx],  
name=None  
) -> Annotated[Any, TV_BroadcastTo_T]
```

```
x = tf.constant([[1, 2, 3]]) # Shape (1, 3,)  
y = tf.broadcast_to(x, [2, 3])  
print(y)  
tf.Tensor(  
[[1 2 3]  
[1 2 3]], shape=(2, 3), dtype=int32)
```

```
x = tf.constant([[1, 2, 3]]) # Shape (1, 3,)
```

```
y = tf.broadcast_to(x, [2, 3])
```

Documentation

Broadcast an array for a compatible shape.

View aliases

Compat aliases for migration

See

Migration guide for
more details.

`tf.compat.v1.broadcast_to`

Used in the notebooks

- Introduction to Tensors
 - Eight schools
 - Human Pose Classification with MoveNet and TensorFlow Lite
 - FFJORD
 - TFP Release Notes notebook (0.13.0)

Broadcasting is the process of making arrays to have compatible shapes for arithmetic operations. Two shapes are compatible if for each dimension pair they are either equal or one of them is one.

For example:

In the above example, the input Tensor with the shape of [1, 3] is broadcasted to output Tensor with shape of [2, 3].

When broadcasting, if a tensor has fewer axes than necessary its shape is padded on the left with ones. So this gives the same result as the previous example:

When doing broadcasted operations such as multiplying a tensor by a scalar, broadcasting (usually) confers some time or space

benefit, as the broadcasted tensor is never materialized.

However, `broadcast_to` does not carry with it any such benefits.

The newly-created tensor takes the full memory of the broadcasted shape. (In a graph context, `broadcast_to` might be fused to subsequent operation and then be optimized away, however.)

Returns